

Maximization of the operational multiplication factor in a photon driven subcritical reactor: a methodological approach for temperature and poison feedback

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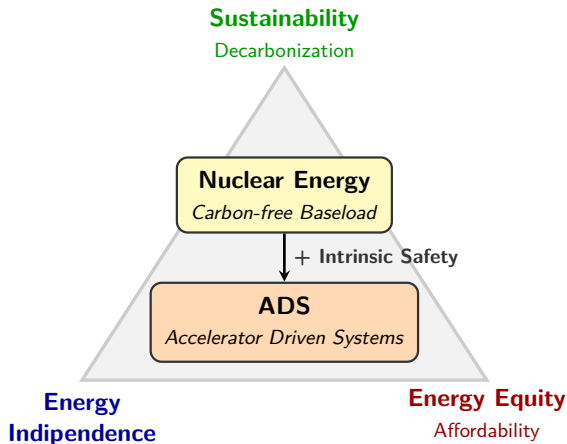
The Global Energy Challenge

Strategic Energy Security

Rising geopolitical tensions highlight the urgent need for **energy independence**. Securing a stable, affordable, and resilient electricity supply is now critical for global economic growth.

Environmental Sustainability

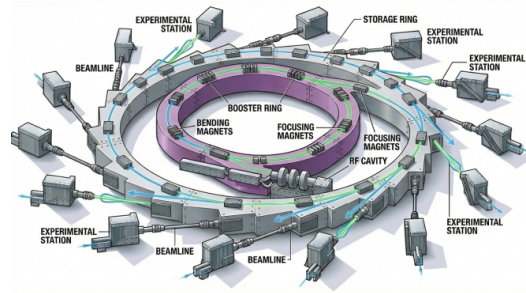
Simultaneously, the transition toward a **carbon-free baseload** is mandatory to drastically reduce global CO₂ emissions without compromising grid stability.



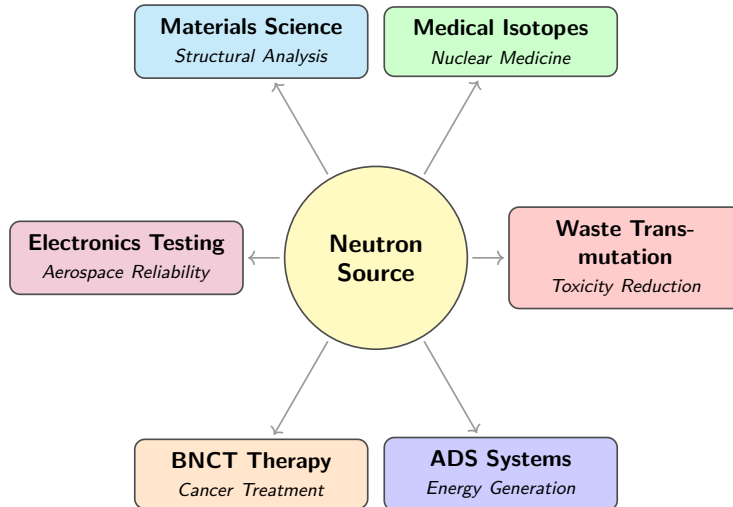
SYNERGY Project: Photonic Source

To overcome the limitations of accelerator based neutron sources — specifically the excessive heat deposition of charged particles on the converter — this thesis is developed within the SYNERGY project:

- Synchrotron-based neutron source utilizing extracted γ -photons (1–37 MeV) to mitigate thermal load on the converter
- Photonuclear (γ, xn) reactions and photofission (γ, f) on a natural Uranium target
- Photonic rate: $S_\gamma = 2.02 \times 10^{17} \text{ } \gamma/\text{s}$
- Neutronic rate up to 10^{15} n/s
- Beam power: $\sim 200 \text{ kW}$ per beamline



Neutron Applications: A Multi-User Platform



Thesis Context and Objective

Subcritical System

A reactor with $k_{eff} < 1$ requires an **external neutron source** to sustain the chain reaction. During operational transients, **feedback** modify k_{eff} .

Objective

Develop a methodology to **maximize** k_{eff}^{HFP} (deeply linked to power production) ensuring that $k_{eff}^{CZP} \leq 1$ under any operating condition.

- **CZP** (Cold Zero Power): $T = 300$ K — reactor shutdown
- **HFP** (Hot Full Power): $T_{fuel} = 600$ K, $T_{mod} = 350$ K — nominal operation

Reactivity Feedback

During the transition from HFP to CZP, the temperature drop induces a **reactivity insertion**. The primary feedback mechanisms are:

Doppler Effect

- Narrowing of ^{238}U capture resonances due to reduced thermal agitation
- Decreased epithermal absorption $\Rightarrow \Delta\rho > 0$

Moderator Effect

- Increase in water density as temperature decreases
- Enhanced neutron thermalization $\Rightarrow \Delta\rho > 0$

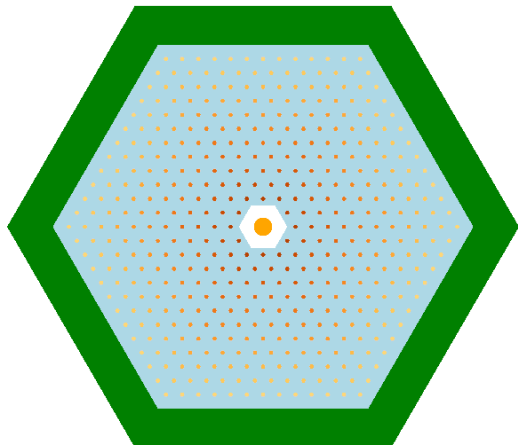
Reactivity Variation

$$\Delta\rho = \frac{k_2 - k_1}{k_1 \cdot k_2} \quad [\text{pcm}]$$

Safety Constraint

$$k_{max}^{HFP} \approx k_{max}^{CZP} - \sum_i \Delta\rho_i$$

Core Topology and Variable Enrichment



Hexagonal Lattice (XY Section)

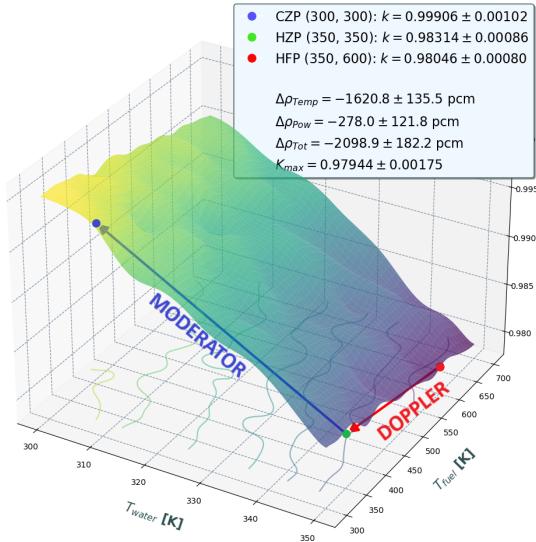
Core Materials

- **Fuel:** UO_2 pins with Zirconium cladding.
- **Moderator:** Light water doped with 15% heavy water ($H_2O + 15\% D_2O$).
- **Reflector:** Nuclear graphite.

Enrichment Strategy (Fuel Zoning)

Radial linear profile: ^{235}U enrichment varies from the inner rings towards the core periphery.

Thermodynamic Multiplication Surface



3D map of $k_{eff}(T_{fuel}, T_{mod})$ in the HFP $\rightarrow$ HZP $\rightarrow$ CZP transient.

The system cooling causes a **reactivity increase** that must be compensated by the initial subcriticality margin.

In addition, another 200 pcm are added as a safety margin to account for uncertainties in the feedback quantification.

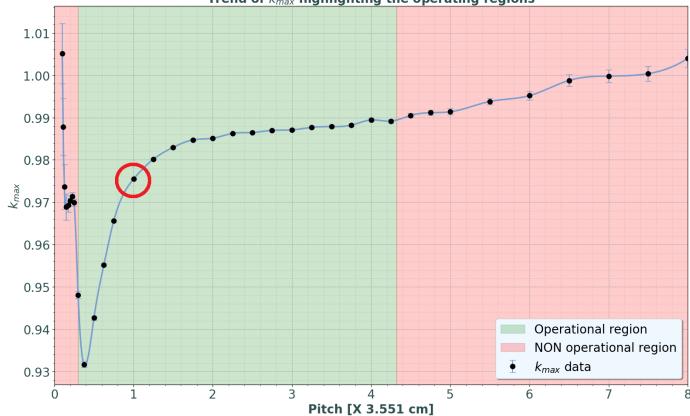
Pitch optimization

Keeping the operating temperatures fixed, what is the optimal pitch to maximize k_{max} ?

Lattice Pitch Optimization

Pitch dictates the moderation level and the neutron spectrum, directly affecting the multiplication factor.

Trend of k_{max} highlighting the operating regions



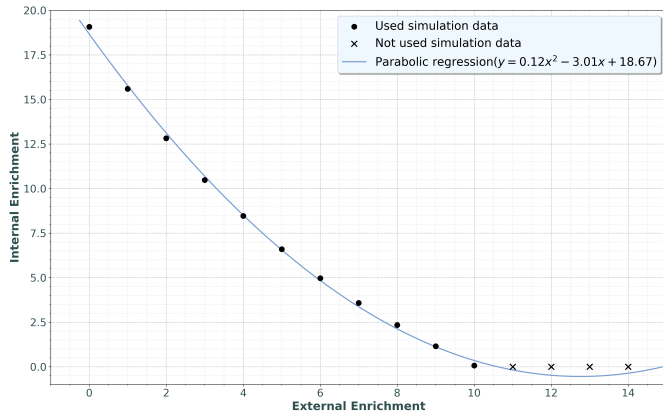
Non-Operational Zone

- Reactivity balance demands an unphysical layout.
- ^{235}U enrichment strictly exceeds the 50% technological limit.

Operational Zone

- **Feasible operation:** Target matched via fuel zoning.
- Radial enrichment finely modulated within safety boundaries.

Stable Solution Space (3.551 cm Pitch)

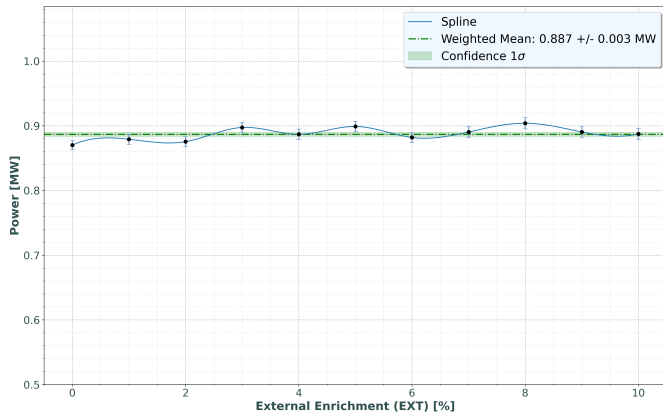


Macroscopic Degeneracy

Multiple radial enrichment profiles yield the same target $k_{\text{HFP}}^{\text{max}}$.

- **Equivalent Solutions:** Stable configurations lie on a continuous parabolic curve.
- **Technological Limit:** Crossed points (×) are discarded as ^{235}U enrichment exceeds the 50% safety threshold.

Extracted Power Invariance

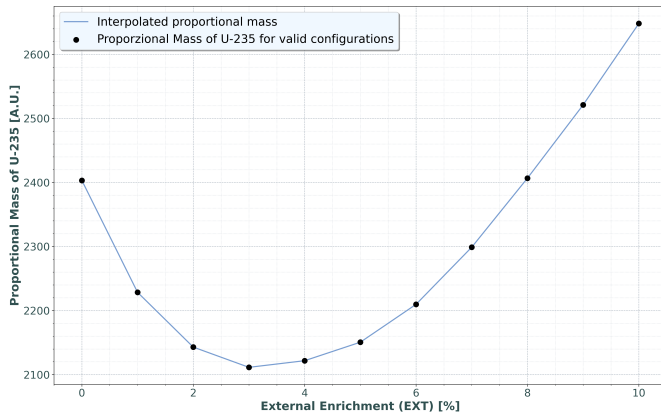


Power Stability

Thermal power generation remains strictly constant across all equivalent enrichment topologies.

- Stable output at $\approx 0.887 \text{ MW}_{\text{th}}$ for a pitch of 3.551 cm.
- Microscopic fluctuations are purely due to **Monte Carlo statistical variance**.
- Formally proves the *thermodynamic consistency* of the model.

Uranium Mass Optimization



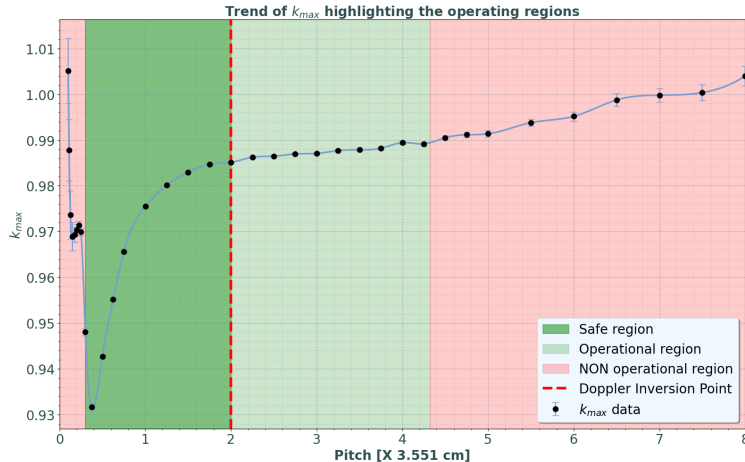
Fuel Economy

Identifying the exact layout that minimizes the required critical mass of Uranium-235.

- The drift shows a distinct **global minimum** (at $\approx 3\%$ external enrichment).
- This optimal point reduces fuel costs and enhances proliferation resistance, while perfectly maintaining the target $k_{HFP}^{\max}$ and power output.

Lattice Pitch Optimization

Pitch dictates the moderation level and the neutron spectrum, directly affecting the multiplication factor.

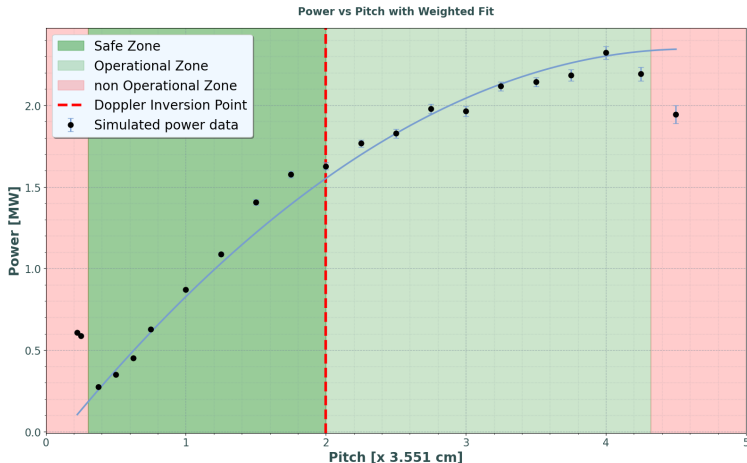


Extreme Over-Moderation

For pitch multipliers > 2.0, operation becomes physically unacceptable:

- The excessive moderator volume reduces the epithermal flux.
- Thermal broadening of ^{238}U resonances loses effectiveness, inverting the feedback sign.
- Core self-stabilization is nullified, risking a thermal runaway.

Results: Power vs. Pitch



Feasible Maximum

- Pitch ≈ 7 cm
- $P_{th} = 1.65$ MW
- **Stable and safe**
- Q -Value = 4.12

$$Q = \frac{P_{Th}}{P_{El}/\eta_{Th-El}}$$

Neutron Poisons: Xe-135 and Sm-149

Other than the thermal feedback, the presence of neutron poisons (fission products with high absorption cross section) can significantly reduce k_{eff} during operation. The main poisons in a uranium-fueled reactor are:

Xenon-135

- $\sigma_a \approx 2 \times 10^6$ b
- Transient: **post-shutdown peak**
- Decays with $T_{1/2} = 9.2$ h

$$N_{Xe}^{eq} = \frac{(\gamma_I + \gamma_{Xe})\Sigma_f\phi}{\lambda_{Xe} + \sigma_a\phi}$$

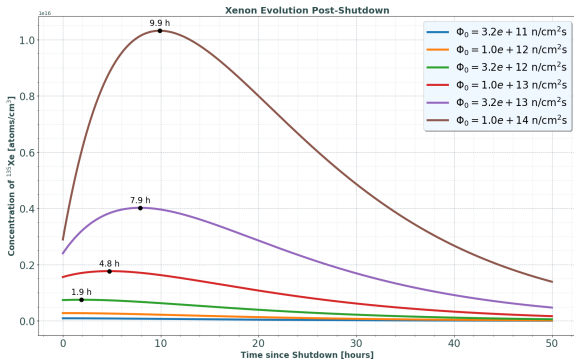
Samarium-149

- $\sigma_a \approx 4.1 \times 10^4$ b
- **Stable** ($\lambda_{Sm} = 0$)
- Permanent asymptotic buildup

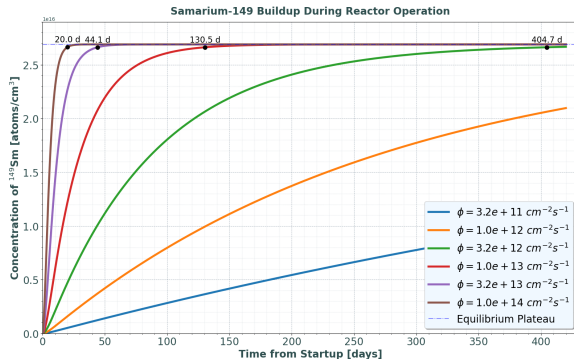
$$N_{Sm}^{eq} = \frac{\gamma_{Pm}\Sigma_f}{\sigma_{a,Sm}}$$

Burnup Simulations: Poison Evolution

From the analytical resolution of the Bateman equations, it is possible to observe the behavior as the flux varies:

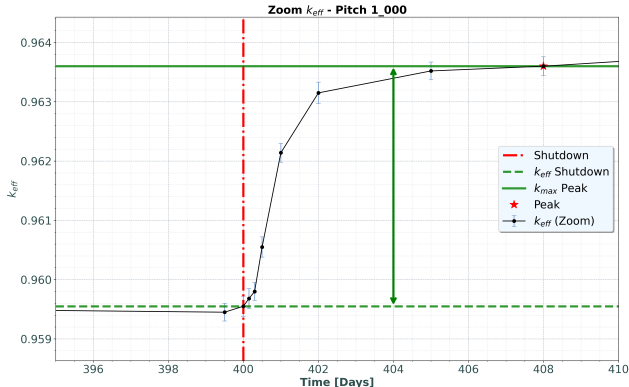


Xe-135: post-shutdown peak



Sm-149: asymptotic plateau

Evolution of k_{eff} over the Life Cycle



The post-shutdown k_{eff} evolution is driven by the net reactivity balance of neutron poisons:

- **Xenon-135** decays entirely without a transient peak, releasing positive reactivity.
- **Samarium-149** asymptotically accumulates, introducing a negative reactivity penalty.

Since the positive reactivity gained from Xenon's disappearance strictly outweighs the Samarium penalty, the system experiences a **net reactivity gain**, leading to the asymptotic growth of the fundamental eigenvalue.

Poisoning Reactivity vs Pitch

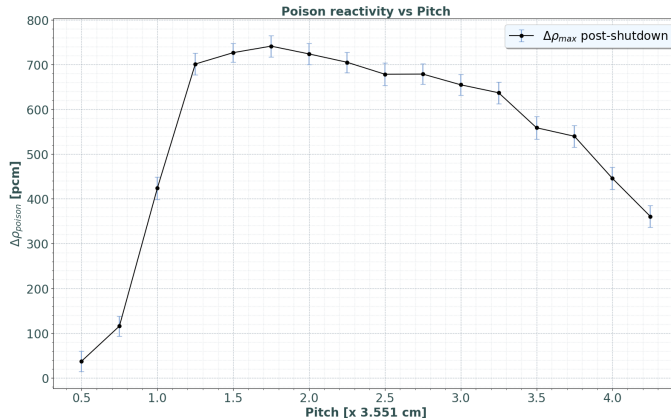


Figure: Evolution of $\Delta\rho_{\text{Poison}}$ as the Pitch varies. The analysis highlights the absence of a global optimum, identifying instead a limiting condition ("worst-case") near the x2 multiplier.

Total Reactivity Balance

$$\Delta\rho_{tot} = \Delta\rho_{Doppler} + \Delta\rho_{Mod} + \Delta\rho_{Poison} + \Delta\rho_{Safety}$$

Lattice Pitch: 3.55 cm

Contribution	[pcm]
Doppler	417
Moderator	1897
Poisons	424
Safety	200
Total	2938

Operational Limit

$$k_{max}^{HFP} \approx 1 - 2938 \text{ pcm} = \boxed{0.97062}$$

Below this limit, safety is guaranteed in any thermal transient.

HFP Operability vs Pitch

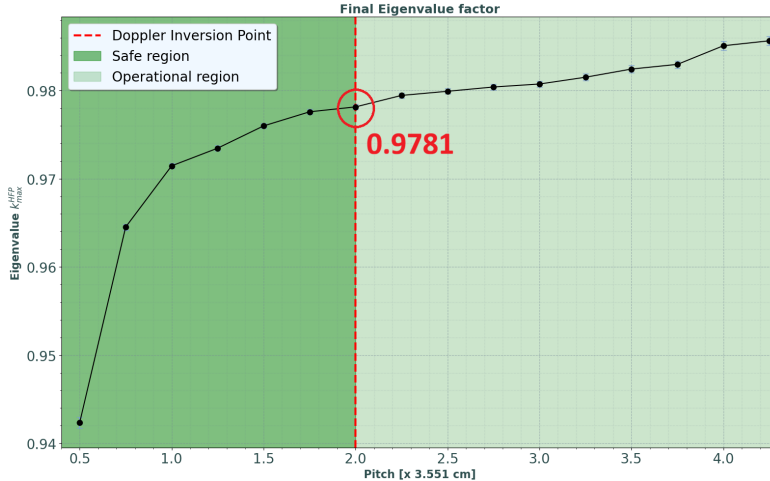


Figure: Maximum k_{eff} value in HFP regime as the pitch varies. The trend shows a constant improvement towards pitch $\times 4$, where the highest available safe multiplication value is achieved.

Conclusions: A Micro-ADS Perspective

Safety & Physics

- **Intrinsic Safety:** $k_{\text{HFP}} = 0.9781$, strictly guaranteeing $k_{\text{CZP}} < 1$ against all feedbacks.
- **Optimal Pitch (7 cm):** Safely prevents the Doppler sign inversion.
- **Performance:** 1.65 MW_{th} output (Q -value = 4.12).

Real-World Applications

A versatile Micro-Reactor acting as a zero-carbon "nuclear heat pump":

- **District & Process Heat:** Localized, zero-carbon thermal energy for industrial use.
- **Water Desalination:** Cogeneration of electricity and clean water.

This analysis can be adopted for different reactor configurations, with the aim of identifying the best scenario for having a robust and localized energy source.

Conclusion

Framework

- **Parametric** and modular optimization procedure.
- **Intrinsically agnostic**: easily adaptable to any alternative reactor layout (geometry, fuel, reflector, or moderator).

Quantitative Output vs. Lattice Pitch

- Maximum allowable operational eigenvalue ($k_{\text{HFP}}^{\text{max}}$).
- Extracted thermal power and *Q-value*.
- Doppler, moderator density, and poisoning reactivity feedback.

Original Contribution

First systematic quantification of the **total reactivity load** in a photon-driven ADS system through fully coupled Monte Carlo and depletion simulations (OpenMC + CRAM).

Future Developments

- **Thermal-Hydraulic Coupling:** 3D spatial integration between OpenMC and CFD/sub-channel solvers for exact local temperature gradient evaluation.
- **Dynamic Beam Control:** Implementation of a feedback control loop on the photon source intensity $S(t)$ to compensate for poison accumulation and maintain a constant thermal power.
- **Source Kinematic Refinement:** Integration of the real anisotropic angular distributions (peaked at 90° for GDR) of the photoneutron emission.
- **Fast Spectrum Extension:** Application of the framework to a fast ADS core (Lead/LBE cooled) to optimize Minor Actinides (MA) transmutation.
- **Optimization:** Multi-objective optimization of the core (fuel, moderator, temperature , ...) to maximize k_{eff} and power output.

Thank you for your attention!

Questions?

Variable Enrichment Geometry

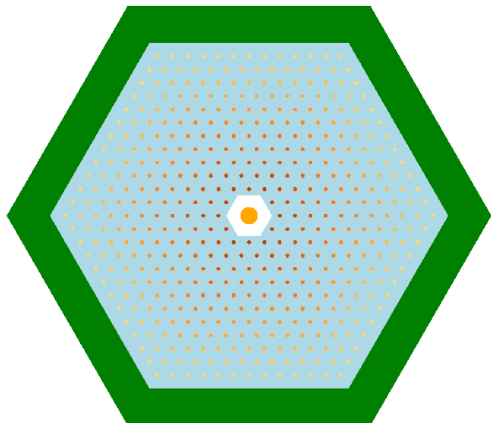


Figure: XY plane section.

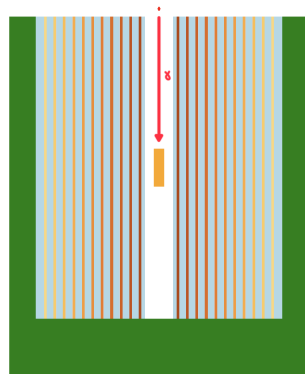


Figure: XZ plane section.

Stable Solution Space

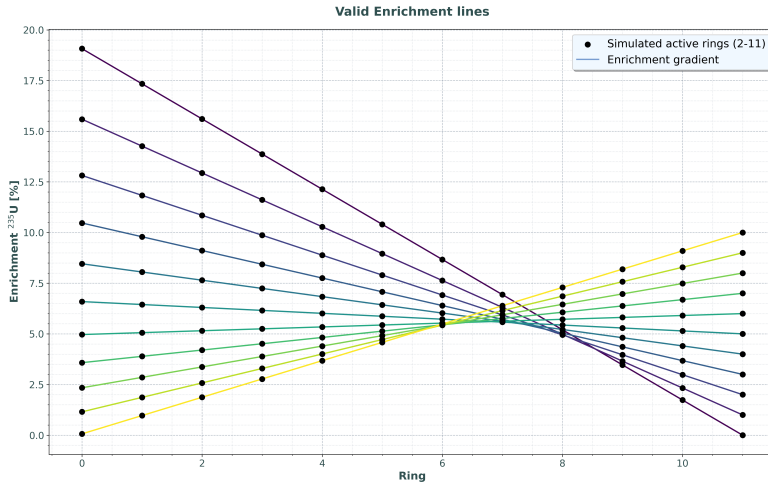
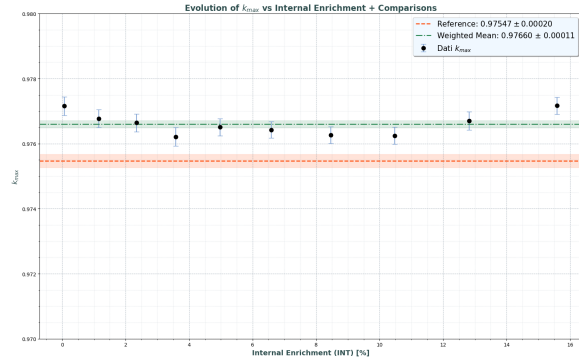


Figure: Radial extrapolation for the enrichment gradient.

Model Consistency and Verification of k_{max}

- simulation across different linear enrichment configurations aimed at matching the reference k_{max} .
- Simulations performed on the three fundamental thermodynamic states: **CZP**, **HZP**, and **HFP**.
- The weighted average analysis confirms the substantial invariance of k_{max} values compared to the limiting case of homogeneous enrichment.
- This behavior formally validates the flexibility of the adopted geometric methodology.



Performance Parameters: The Q-Factor

In an ADS system, the energy gain factor Q describes the amplification of the source power within the subcritical core:

$$Q = \frac{P_{thermal}}{P_{beam}} = \psi \frac{k_{eff}}{1 - k_{eff}}$$

Where ψ represents the relative importance factor of source neutrons.

For a consistent evaluation, a system Q -factor (Q_{sys}) is defined with respect to the electrical power of the synchrotron:

$$Q_{sys} = \frac{P_{thermal}}{P_{el, sync} / \eta_{th}}$$

Technology	Q-Value (Target/Record)	Regime
ITER	~ 10	Fusion (Under construction)
SPARC	> 11	Fusion (Under development)
JET	0.67	Fusion (1997 Record)

Depletion Simulations (Burn-up)

Transport-Isotopic Evolution Coupling

The simulation in *depletion* mode allows tracking the time evolution of the nuclear fuel isotopic composition under the effect of the neutron flux, by solving the coupled Monte Carlo transport and Bateman equations.

- **Transport Phase (OpenMC):** Calculation of the spatial and spectral neutron flux $\Phi(\vec{r}, E)$ and microscopic reaction rates $(\sigma_i \phi)$ at the beginning of the time-step.
- **Evolution Phase (Bateman):** Use of the extracted rates to numerically solve the system of linearized differential equations, updating the atomic densities $N_i(t)$ for the next step.
- **Composition Feedback:** The accumulation of fission products (poisons) and the consumption of fissile material alter the macroscopic properties of the medium (Σ_a, Σ_f), modifying the neutron response in subsequent steps.

Time Evolution of ^{135}I and ^{135}Xe

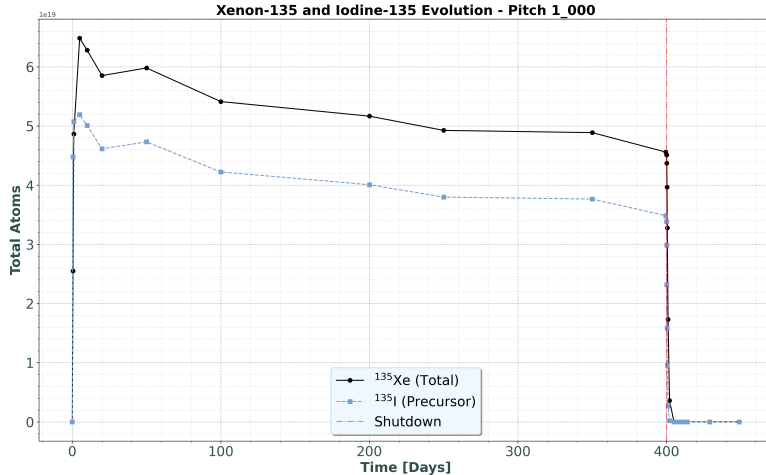


Figure: Coupled behavior of Iodine-135 and Xenon-135 atomic densities during nominal operation and during the post-shutdown phase.

Time Evolution of ^{135}I and ^{135}Xe

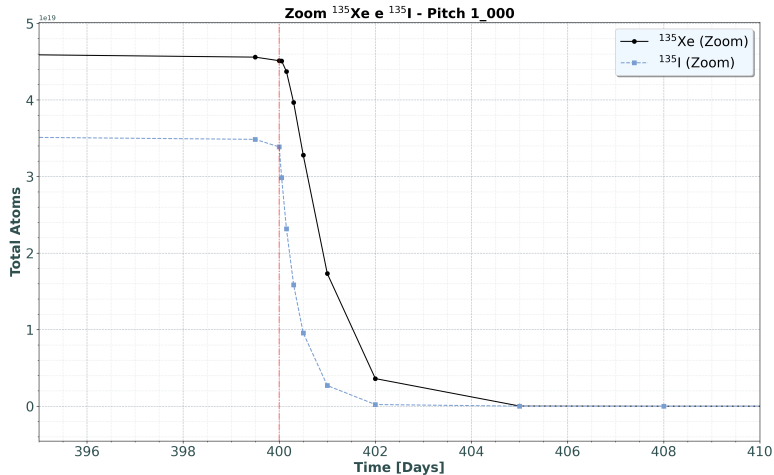


Figure: The graph highlights the onset of the post-shutdown poisoning peak induced by the decoupling of the half-lives of the two nuclides.

Time Evolution of ^{149}Sm

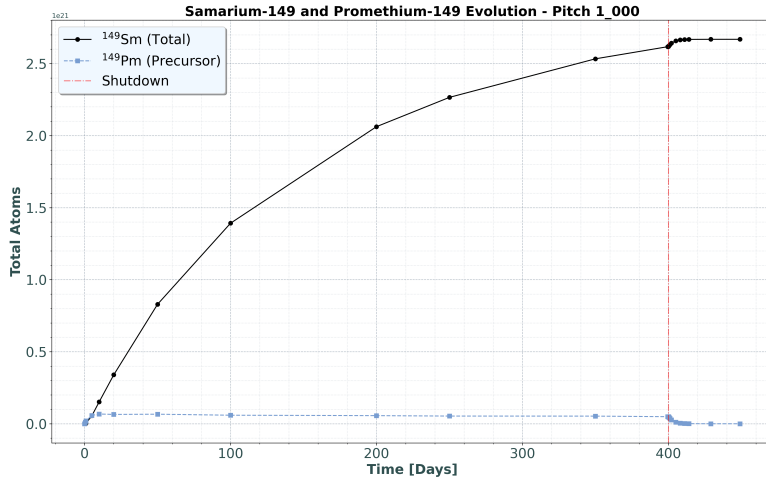


Figure: The graph highlights the asymptotic behaviour of samarium during core activity.

Core Thermal Power Evolution

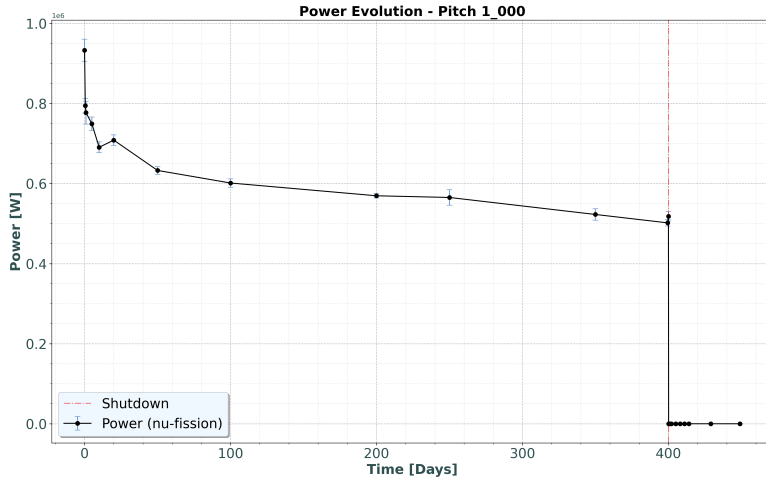


Figure: Time profile of the overall thermal power expressed in W over the fuel lifecycle.

Evolution of the Effective Multiplication Factor (k_{eff})

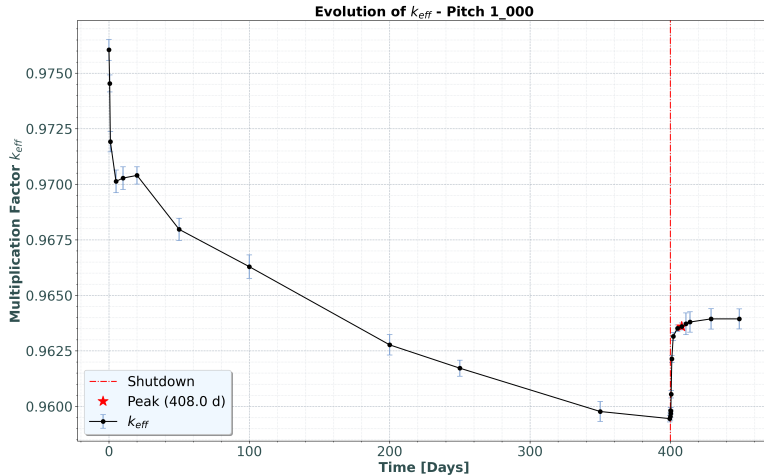


Figure: Trend of the effective multiplication factor k_{eff} as a function of Hot Full Power (HFP) operating days. The plot documents the legal safety margins and the overall stability of the system with respect to poisoning and fissile

Mathematical Foundation: The Transport Equation

The steady-state neutron balance is governed by the transport equation:

Operator Formulation

$$\mathcal{M}\Phi(\vec{r}, E, \hat{\Omega}) = \frac{1}{k_{eff}} \mathcal{F}\Phi(\vec{r}, E, \hat{\Omega})$$

The two operators describe the competing physical rates in the core:

- **$\mathcal{M}$ (Migration/Destruction Operator):** Quantifies net losses. It includes geometric leakage (*streaming*, $\hat{\Omega} \cdot \nabla \Phi$), total interactions ($\Sigma_t \Phi$), and kinematic repopulation due to scattering (*in-scattering*).
- **$\mathcal{F}$ (Production Operator):** Quantifies the steady-state source, describing the emission of new prompt neutrons induced by the integral fission rate ($\chi \nu \Sigma_f \Phi$).

Mathematical Foundation: The Krein-Rutman Theorem

Rewriting the equation as $\mathcal{M}^{-1}\mathcal{F}\Phi = k_{eff}\Phi$, the operator $\mathcal{M}^{-1}\mathcal{F}$ maps one neutron generation to the next. Since atomic densities and cross-sections are strictly positive quantities, this operator is **rigorously positive**.

Eigenvalue Dominance (k_{eff})

By the **Krein-Rutman Theorem**, a positive integral operator guarantees the existence of a fundamental eigenvalue ($k_0 \equiv k_{eff}$) that is real, positive, and **larger in magnitude** than all other eigenvalues in the spectrum ($|k_0| > |k_n|$).

Physical Meaning of the Eigenfunction (Φ_0)

Only the dominant eigenvalue k_{eff} is associated with an eigenfunction Φ_0 that is **everywhere positive**. Higher-order harmonics inevitably exhibit nodes (negative fluxes, which are physically impossible in steady-state conditions).